



BITS Pilani

Pilani Campus

Deep Neural Network

Dr. S. Suresh Kumar

Deep Neural Network



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- I here by acknowledge all the contributors for their material and inputs.
- I have provided source information wherever necessary
- I have added and modified the content to suit the requirements of the course



Course Logistics



What we Learn.... (Module Structure)

1. Fundamentals of Neural Network
2. Multilayer Perceptron
3. Deep Feedforward Neural Network
4. Improve the DNN performance by Optimization and Regularization
5. Convolutional Neural Networks
6. Sequence Models
7. Attention Mechanism
8. Neural Network Search
9. Time series Modelling and Forecasting
10. Other Learning Techniques

Books



Textbook

Dive into Deep Learning by Aston Zhang, Zack C. Lipton, Mu Li, Alex J. Smola. https://d2l.ai/chapter_introduction/index.html

Reference book

Deep Learning by Ian Goodfellow, Yoshua Bengio, Aaron Courville <https://www.deeplearningbook.org/>

Evaluation Components and Schedule



	Evaluation Component	Mode	Duration	Weightage	Date and Time
EC1	Quiz I*	Online	15 Mins	5%	Refer to LMS
	Quiz II*		15 Mins	5%	
	Assignment 01		2 Weeks	10%	
	Assignment 02		2 Weeks	10%	
EC2	Mid-sem exam (Closed book)		2 Hours	30%	
EC3	Comprehensive exam (Open book)		2.5 Hours	40%	

* Best of Two



Lab Sessions

- L1 Introduction to PyTorch
- L2 Deep Neural Network with Back-propagation and optimization
- L3 CNN
- L4 RNN
- L5 LSTM
- L6 Auto-encoders

Course Logistics



- Refer Taxila for the following
Handout
Schedule for Webinar
Schedule of Quiz, and Assignments.
Evaluation scheme
Session Slide Deck
Demo Lab Sheets
Quiz-I, Quiz-II
Assignment-I, Assignment-II
Sample QPs
- Lecture Recordings
Available on Microsoft teams

Honour Code



All submissions for graded components must be the result of your original effort. It is strictly prohibited to copy and paste verbatim from any sources, whether online or from your peers. The use of unauthorized sources or materials, as well as collusion or unauthorized collaboration to gain an unfair advantage, is also strictly prohibited. Please note that we will not distinguish between the person sharing their resources and the one receiving them for plagiarism, and the consequences will apply to both parties equally.

In cases where suspicious circumstances arise, such as identical verbatim answers or a significant overlap of unreasonable similarities in a set of submissions, will be investigated, and severe punishments⁹ will be imposed on all those found guilty of plagiarism.

Types of Machine Learning Algorithms



Criteria	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Definition	The machine learns by using labeled data	The machine is trained on unlabeled data without any guidance	An agent interacts with its environment by performing actions & learning from errors or rewards
Type of problems	Regression & classification	Association & clustering	Reward-based
Type of data	Labeled data	Unlabeled data	No predefined data
Training	External supervision	No supervision	No supervision
Approach	Maps the labeled inputs to the known outputs	Understands patterns & discovers the output	Follows the trial-and-error method



AI - ML - DL

AI : Artificial intelligence is the science of making things smart. The aim is make machines perform human tasks. Eg: Robot cleaning a room.

ML : Machine learning is an approach to AI. The machine learns or perform tasks through learning by experience.

DL : Deep Learning is a technique for implementing machine learning to recognise patterns.



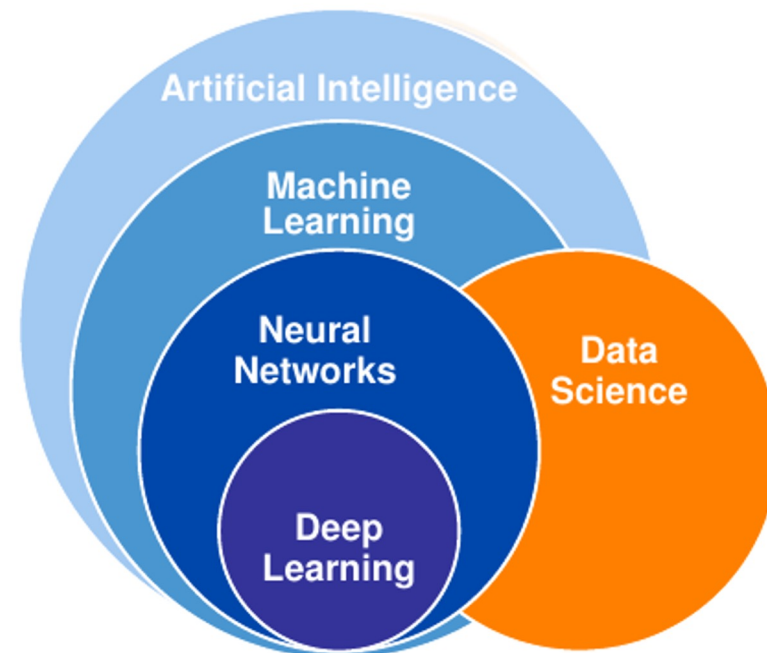
Deep (Machine) Learning

- Deep learning is a specific subfield of machine learning.
- Learning representations from data that puts an emphasis on learning successive layers of increasingly meaningful representations.
- The **deep** in deep learning stands for this idea of successive layers of representations.
- The number of layers that contribute to model the data is called the **depth** of the model.
- In deep learning, the layered representations are learned via models called **neural networks**, structured in literal layers stacked on top of each other.

Where in AI sits DL?



- **Deep Learning:** neural network with **three or more layers**
- AI is a general field that encompasses machine learning and deep learning, but that also includes many more approaches that don't involve any learning.





What is Deep Learning?

Definitions of Deep Learning



- Deep Learning is a type of machine learning based on artificial neural networks in which multiple layers of processing are used to extract progressively higher-level features from data.
- Deep learning is a method in artificial intelligence (AI) that teaches computers to process data in a way that is inspired by the human brain.
- Deep learning is a machine learning technique that teaches computers to do what comes naturally to humans: learn by example.
- Deep learning is a subset of machine learning, which is essentially a neural network with three or more layers.
- Deep Learning gets its name from the fact that we add more "Layers" to learn from the data.



Why Deep Learning?

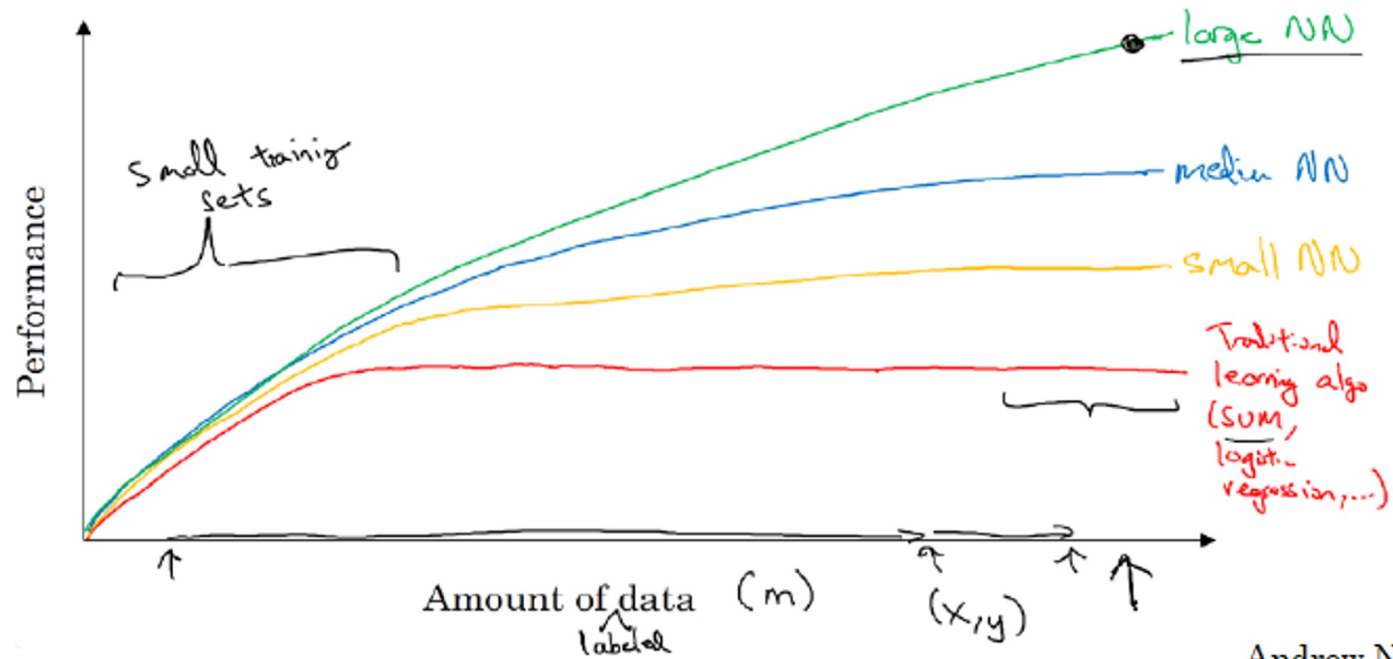


Why Deep Learning?

- Large amounts of data
- Lots and lots of unstructured data like images, text, audio, video
- Cheap, high-quality sensors
- Cheap computation - CPU, GPU, Distributed clusters
- Cheap data storage
- Learn by examples
- Automated feature generation
- Better learning capabilities
- Scalability
- Advance analytics can be applied

Why deep learning?

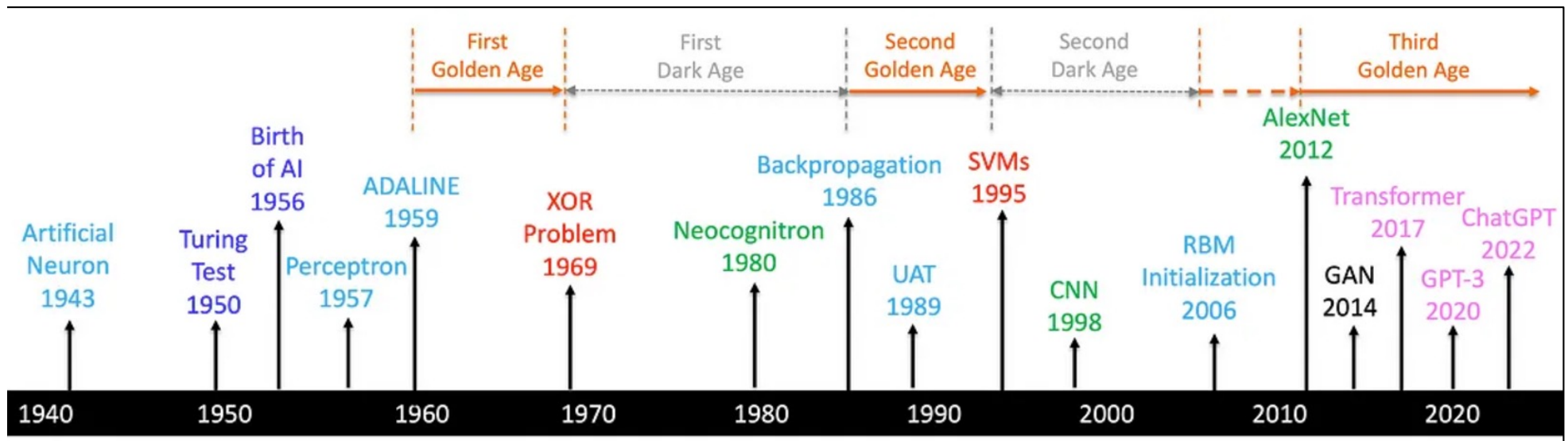
Scale drives deep learning progress



Andrew Ng



Deep Learning Timeline





Applications of Deep Learning

Breakthroughs with neural networks



www.technewsworld.com/story/84013.html

top 40 maps that explain Amazon Web Services Primers | Math n Pro: deeplearning.net/tuto Deep Learning Tutori deep learning PHILIPS - Golden Ears Language Technolog MyIDCare - Dashboa Other bookmarks

TECHNEWSWORLD EMERGING TECH

Computing Internet IT Mobile Tech Reviews Security Technology Tech Blog Reader Services

Microsoft AI Beats Humans at Speech Recognition

By Richard Adhikari
Oct 20, 2016 11:40 AM PT

Print Email

Google+ 5
Tweet 25
Share 45
LinkedIn Share 11
Share 0
share 104

Artificial Intelligence
Reasoning
Science
Computer
Knowledge
Technology
Learning

AI

Image: Adobe Stock

Microsoft's Artificial Intelligence and Research Unit earlier this week reported that its speech recognition technology had surpassed the performance of human transcriptionists.

How do you feel about Black Friday and Cyber Monday?

- They're great -- I get a lot of bargains!
- The deals are too spread out -- I'd prefer just one day.
- They're a fun way to kick off the holiday season.
- I don't like the commercialization of Thanksgiving Day.
- They're crucial for the retail industry and the economy.
- The deals typically aren't that good.

Vote to See Results

E-Commerce Times

Black Friday Shoppers Hungry for New Experiences, New Tech

Pay TV's Newest Innovation: Giving Users Control

Apple Celebrates Itself in \$300 Coffee Table Tome

AWS Enjoys Top Perch in IaaS, PaaS Markets

US Comptroller Gears Up for Blockchain and

Breakthrough with neural networks

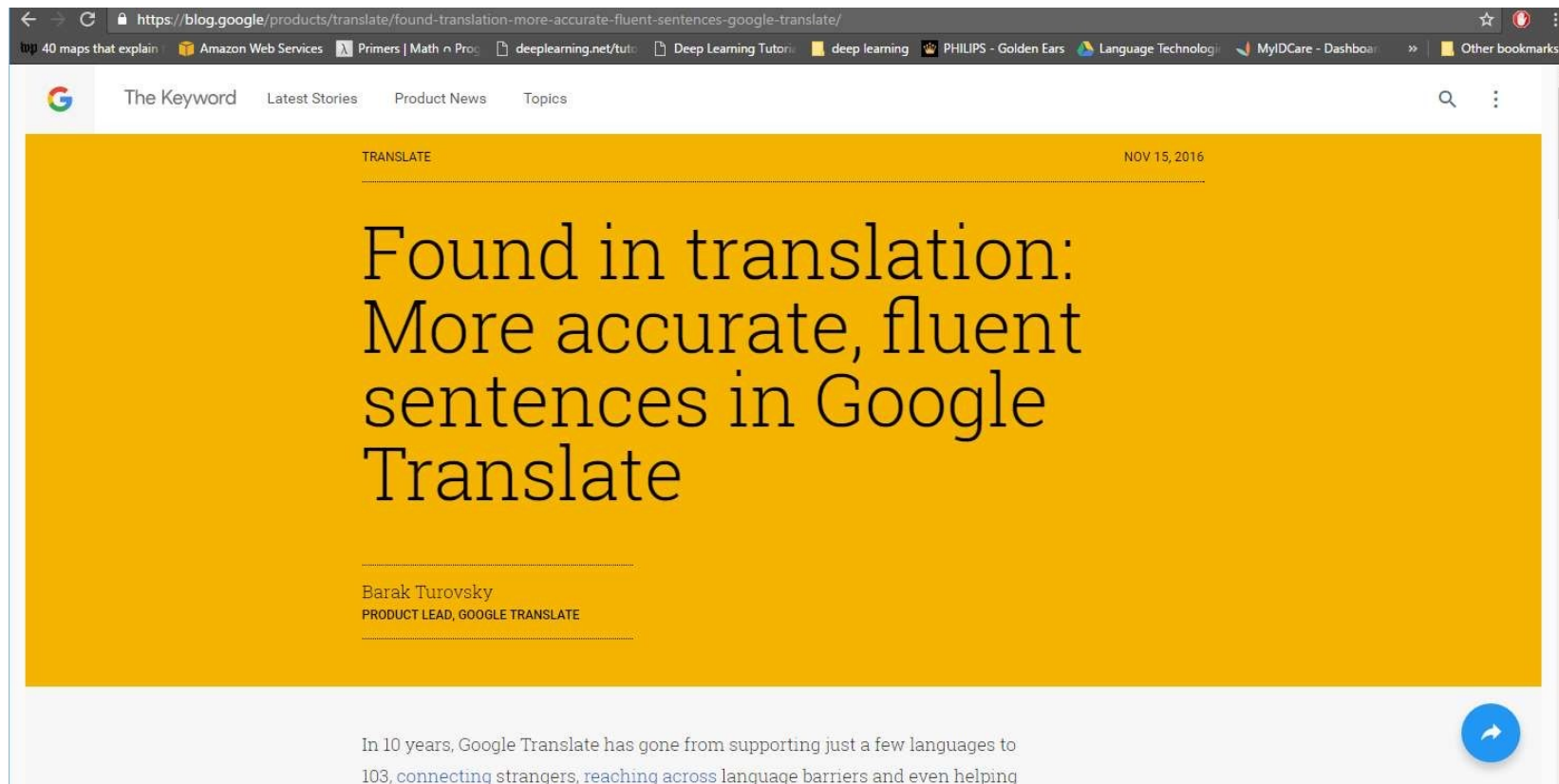
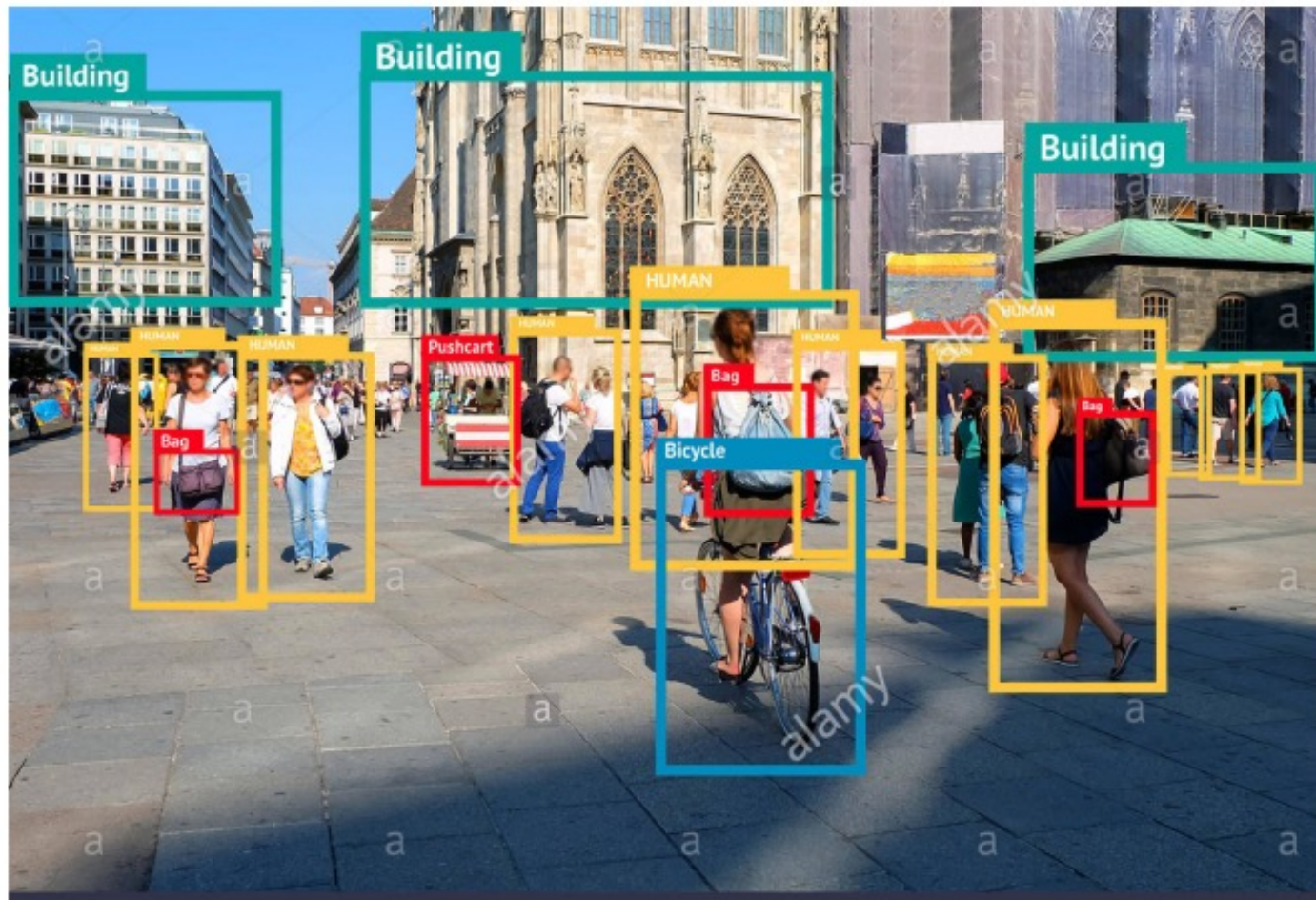
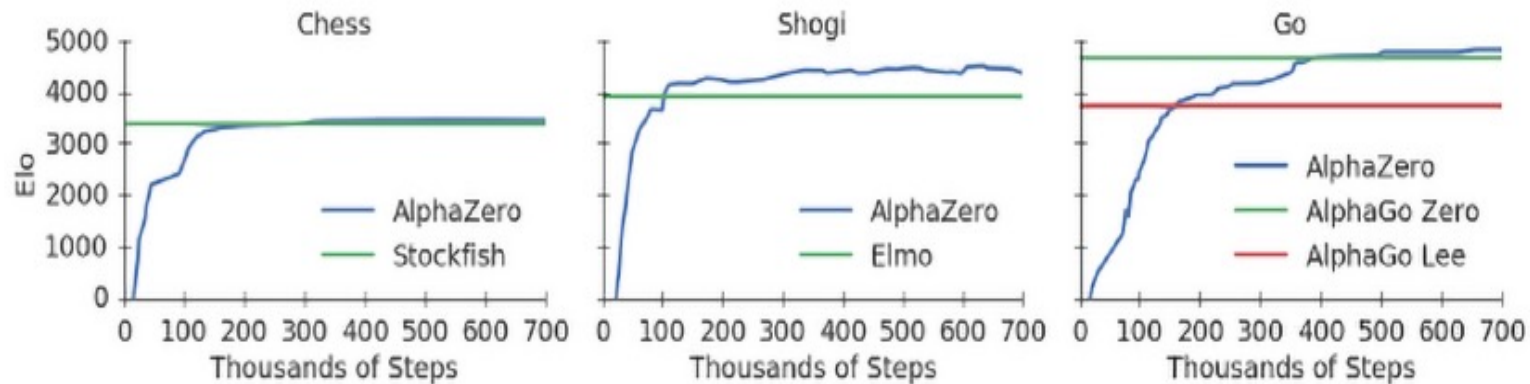


Image segmentation and recognition



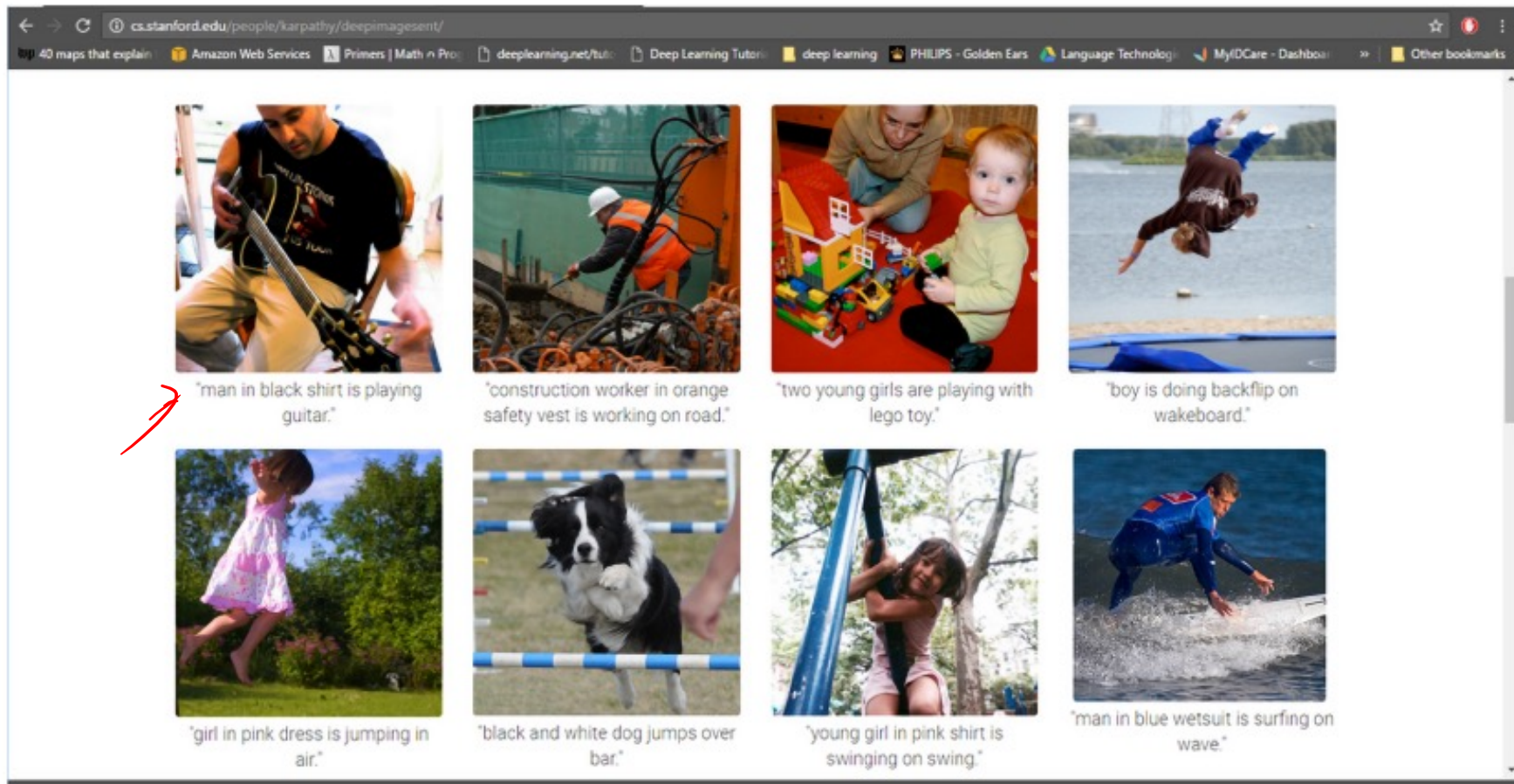


Breakthroughs with neural networks



- Google DeepMind's AI system, **AlphaGo**, learned to master the ancient Chinese game of Go — complex board game of strategy,
- AlphaGo –In October 2015, the original AlphaGo became the first computer Go program to beat a human professional Go player
- AlphaGo Zero - completely self-taught without learning from human games.
- AlphaGo Zero – Played Chess and Shogi

Success with neuralnetworks



The result of an image captioning task from 2015

Breakthroughs with neural networks

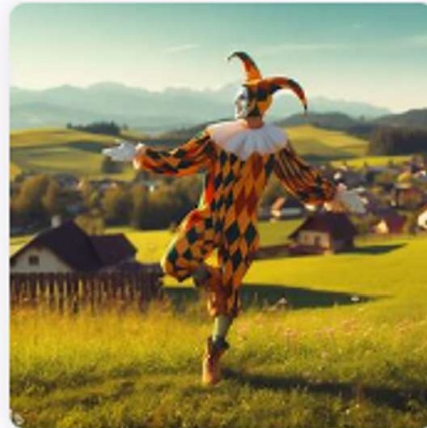
Stable Diffusion

image

ChatGPT

Text
↓ ⇒

why does winching up an anchor raise the anchor instead of pulling the ship down



))) "A harlequin dancing in a field on a sunny beautiful day in a beau..."

Bing, 6:10 PM

Winching up an anchor raises the anchor instead of pulling the ship down because of several factors, such as:

- The design of the windlass, which is the device that operates the anchor cable. The windlass is not meant to haul the ship, but only to lift the anchor and reduce the tension in the cable. The windlass motor is the "weak link" in the anchoring system and can be damaged by excessive force ¹. Therefore, the windlass is used carefully and slowly, and the ship is maneuvered towards the anchor to assist the recovery process ².
- The weight of the anchor and the cable, which are much smaller than the weight of the ship. The anchor and the cable together may weigh a few tons, while a typical ship may weigh a few hundred or thousand tons. Therefore, the force exerted by the windlass on the anchor is not enough to overcome the inertia and buoyancy of the ship.
- The friction between the anchor and the sea bottom, which resists the pull of the windlass. The anchor works by digging into the sea bottom with its fluke or pointed projection, creating a strong grip. To break out the anchor, the chain has to be locked in a chain stopper and the ship has to move in a way that pulls out the anchor from the sea bottom ². If the anchor is stuck or buried too deep, it may be difficult to retrieve it with the windlass alone ³.
- The angle of the cable, which affects the direction of the force applied by the windlass. The cable should be vertical or "up and down" when heaving up the anchor, so that the force is directed upwards. If the cable is inclined or horizontal, it means that the ship is too far from the anchor and that the force is

Applications of Deep Learning

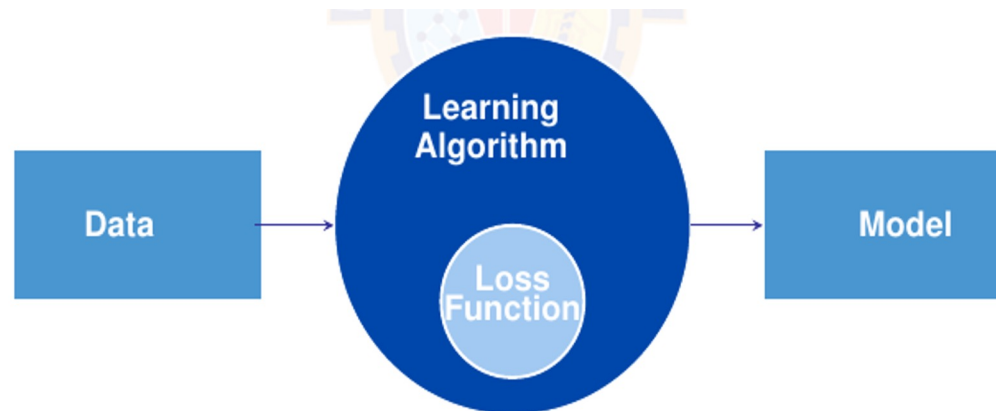
Application	Input	Output	Neural Network
Real Estate	House features	House Price	Std NN
Photo Tagging	Image	Text	CNN
Object detection	Image	Bounding box	CNN
Speech Recognition	Audio	Text transcript	RNN
Translation	English Text	French Text	RNN
Autonomous driving	Image, Sensors Radars	Position of other cars, objects, signals	Hybrid NN



Key components of DL problem

Core components of DL problem

1. The **data** that we can learn from.
2. A **model** of how to transform the data.
3. An **objective function** that quantifies how well (or badly) the model is doing.
4. An **algorithm** to adjust the model's parameters to optimize the objective function.





1. Data

- Collection of examples.
- Data has to be converted to an useful and a suitable numerical **representation**.
- Each example (or data point, data instance, sample) typically consists of a set of attributes called **features** (or covariates), from which the model must make its predictions.
- In the supervised learning problems, the attribute to be predicted is designated as the **label** (or target).
- Mathematically, a set of m examples,
- We need **right** data.

$$Data = \mathcal{D} = \{X, t\}$$

1. Data



- **Dimensionality** of data

- Each example has the same number of numerical values. This data consist of fixed-length vectors. Eg: Image
- The constant length of the vectors as the dimensionality of the data.
- Text data has varying-length data.

2. Model

- Model denotes the **computational machinery** for ingesting data of one type, and spitting out predictions of a possibly different type.
- Deep learning models consist of many successive transformations of the data that are chained together top to bottom, thus the name deep learning.

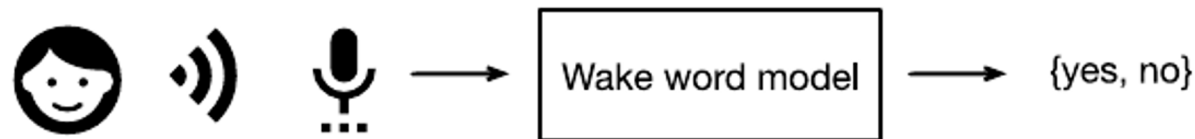


Fig. 1.1.1: Identify a wake word.



3. Objective Function

- **Learning means improving at some task over time.**
- A formal mathematical system of learning machines is defined using formal measures of how good (or bad) the models are. These formal measures are called as **objective functions**.
- By convention, objective functions are defined so that lower is better.
- Because lower is better, these functions are sometimes called **loss functions**.



3. Loss Functions

- To predict numerical values (regression), the most common loss function is squared error.
- For classification, the most common objective is to minimize error rate, i.e., the fraction of examples on which our predictions disagree with the ground truth.



3. Loss Functions

- Loss function is defined **with respect to the model parameters** and **depends upon the dataset**.
- We learn the best values of our model parameters by **minimizing the loss** incurred on a set consisting of some number of examples collected for training. However, doing well on the training data does not guarantee that we will do well on unseen data. I.e **Model has to generalize better**.
- When a model performs well on the training set but fails to generalize to unseen data, we say that it is **overfitting**.



4. Optimization Algorithms

- Optimization Algorithm is an algorithm capable of **searching for the best possible parameters** for **minimizing the loss function**.
- Popular optimization algorithms for deep learning are based on an approach called **gradient descent**.

Example of the Framework

- We have to tell a computer explicitly how to map from inputs to outputs.
- We have to define the problem precisely, pinning down the exact nature of the inputs and outputs, and choosing an appropriate model family.
- Collect a huge dataset containing examples of audio and label those that do and that do not contain the wake word.

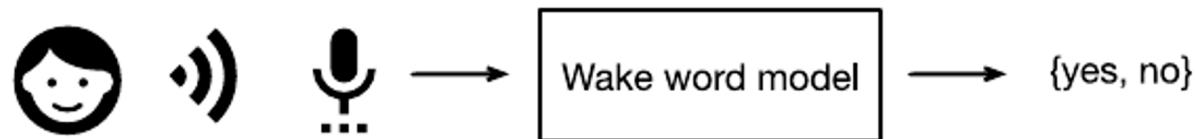


Fig. 1.1.1: Identify a wake word.



Example of the Framework

- Create a Model
 - Define a flexible program whose behavior is determined by a number of parameters.
 - To determine the best possible set of parameters, use the data. The parameters should improve the performance of the program with respect to some measure of performance on the task of interest.
 - After fixing the parameters, we call the program a model.
 - Eg: The model receives a snippet of audio as input, and the model generates a selection among yes, no as output.
 - The set of all distinct programs (input-output mappings) that we can produce just by manipulating the parameters is called a family of models.
 - Eg: We expect that the same model family should be suitable for "Alexa" recognition and "Hey Siri" recognition because they seem, intuitively, to be similar tasks.

Example of the Framework

- The meta-program that uses our dataset to choose the parameters is called a learning algorithm.
- In machine learning, the learning is the process by which we discover the right setting of the parameter coercing the desired behavior from our model.
- Train the model with data.

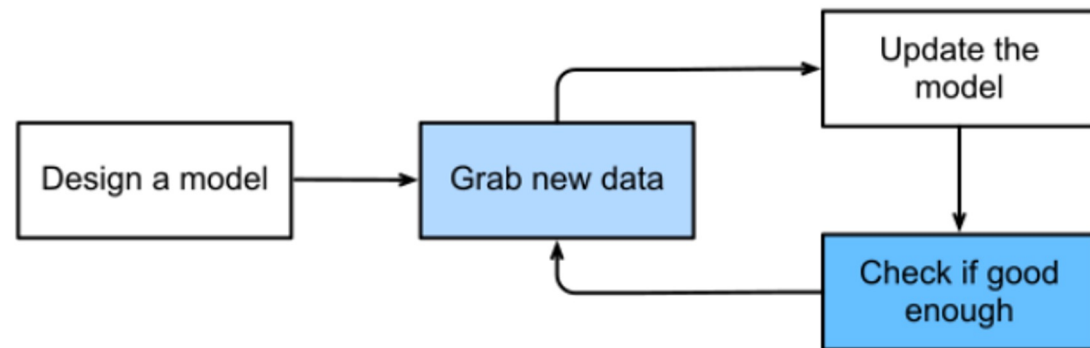


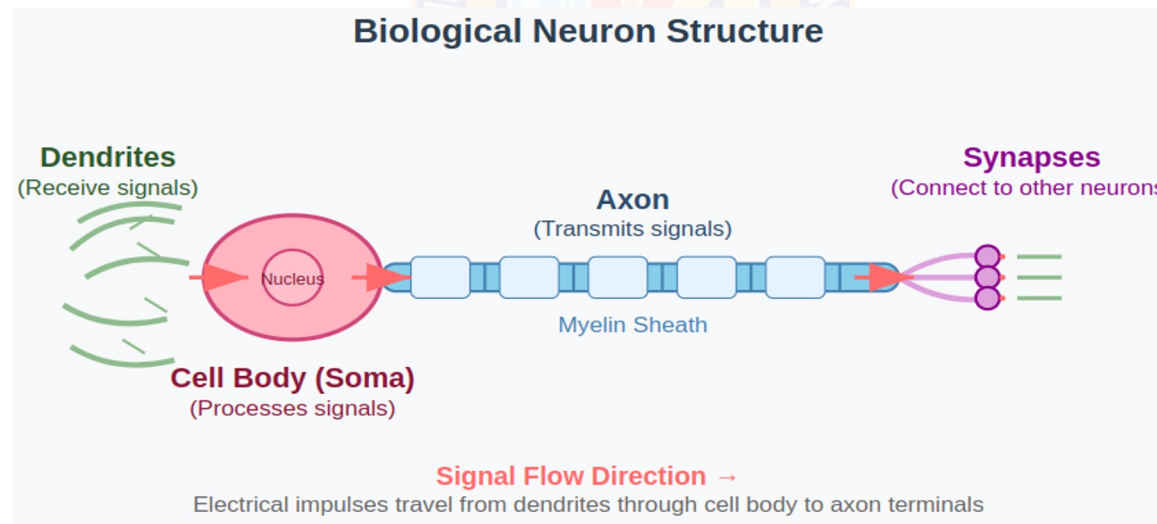
Fig. 1.1.2: A typical training process.

Biological Neuron

lead

Key Components:

- **Dendrites:** Receive signals from other neurons
- **Cell Body (Soma):** Processes incoming signals
- **Axon:** Transmits output signal
- **Synapses:** Connection points between neurons





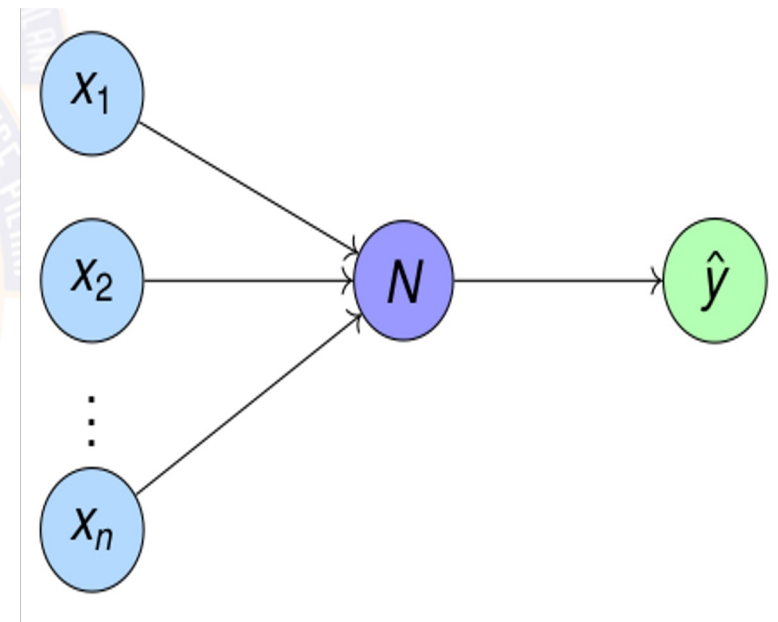
Artificial Neuron



What are Artificial Neurons?

lead

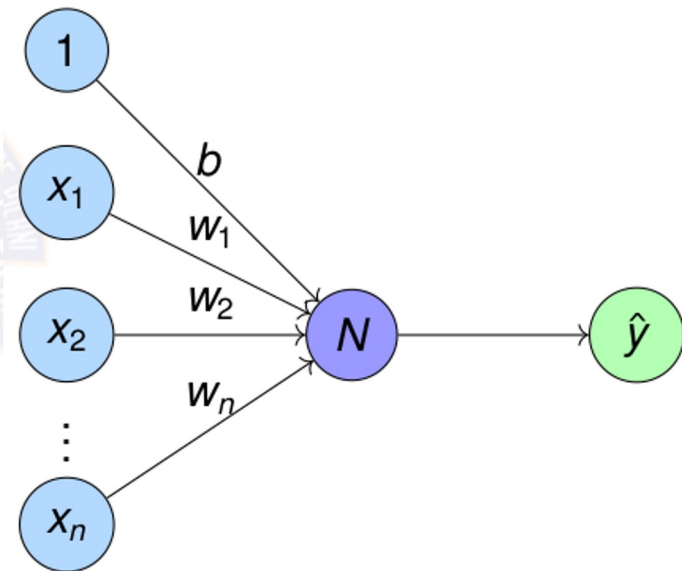
- Neuron is a processing element inspired by how the brain works.
- Similar to biological neuron, each artificial neuron will be do some computation. Each neuron is interconnected to other neurons.
- Similar to brain, the interconnections between neurons store the knowledge it learns. The knowledge is stored as parameters.



Artificial Neuron N

lead

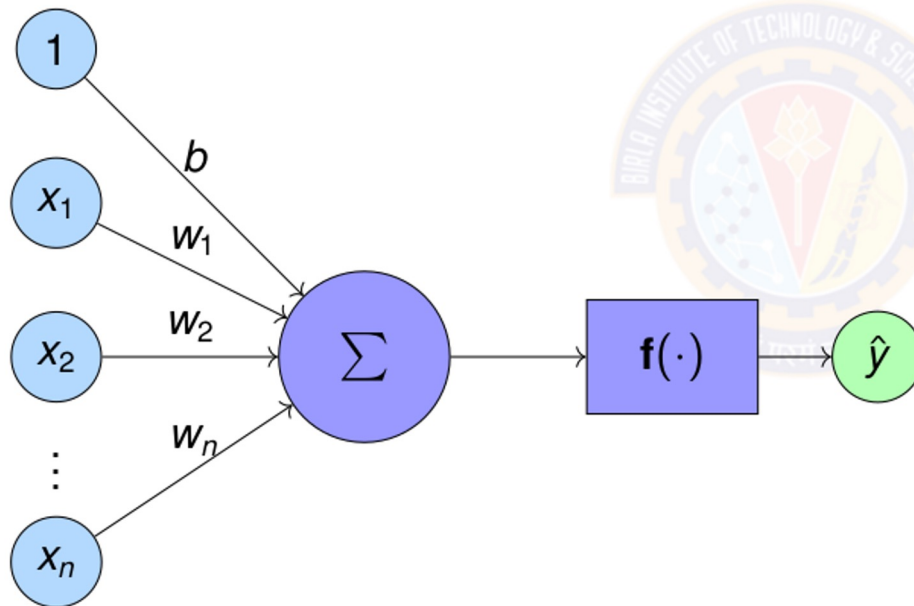
- Processing neuron N performs a very basic computation.
- For each feature x_i , weight w_i shows the importance to the feature
- N generates one numerical output $\hat{y}$



Artificial Neuron – Mathematical Computation

lead

- Sum of weights w_i times corresponding features x_i
- Apply activation function $f()$ on the sum.



$$z = \sum_{i=1}^n w_i x_i + b \quad (1)$$

$$\hat{y} = f \left(\sum_{i=1}^n w_i x_i + b \right) \quad (2)$$

Comparison: Biological vs Artificial Neurons

lead

Biological Neuron

- Complex biochemical processes
- Analog signal processing
- Adaptive synaptic strengths
- Parallel processing
- Learning through experience
- Fault tolerant

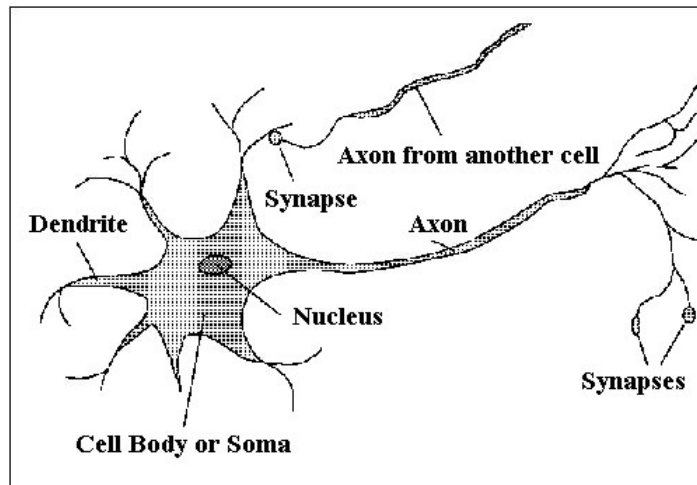
Artificial Neuron

- Simple mathematical model
- Digital computation
- Adjustable weights
- Parallel computation possible
- Learning through algorithms
- Deterministic behavior



Biological Neuron Vs Artificial Neuron

The Human brain - Interconnected Neurons



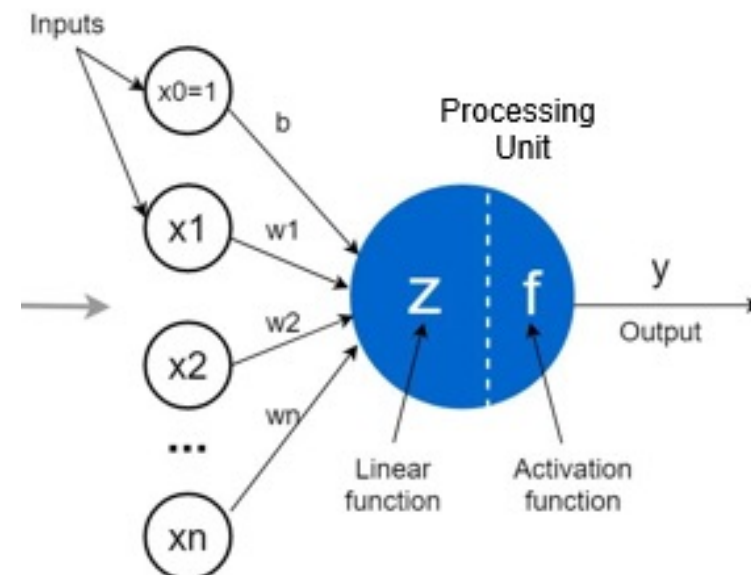
Dendrite: Receives signals from other neurons

Soma: Processes the information

Axon: Transmits the output of this neuron

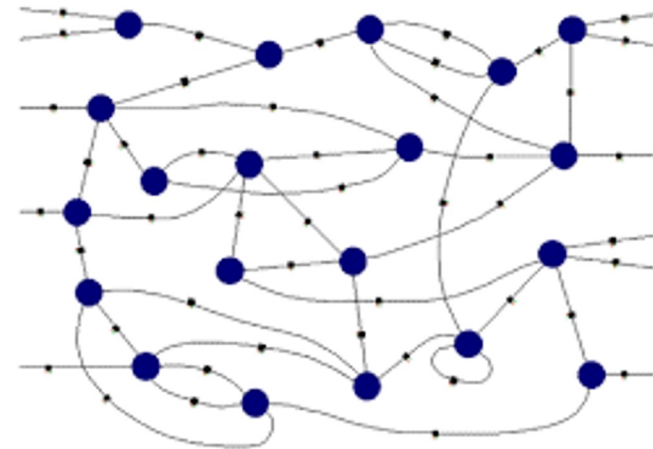
Synapse: Point of connection to other neurons

Biological Neuron	Artificial Neuron
Cell Nucleus (Soma)	Node
Dendrites	Input
Synapse	Weights or interconnections
Axon	Output



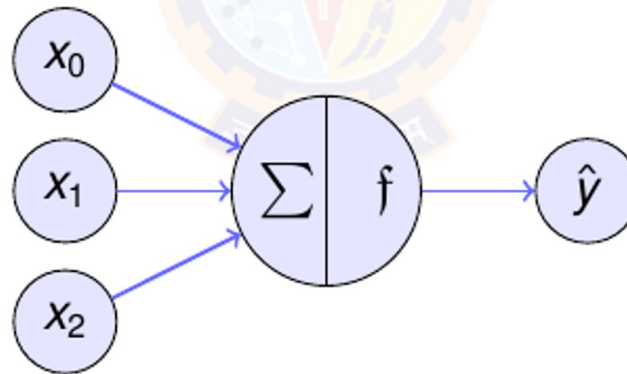
Connectionist Machines

- Network of processing elements, called artificial neural unit.
- The neurons are interconnected to form a network.
- All world knowledge is stored in the connections between these elements.
- Neural networks are connectionist machines.



What are Artificial Neurons?

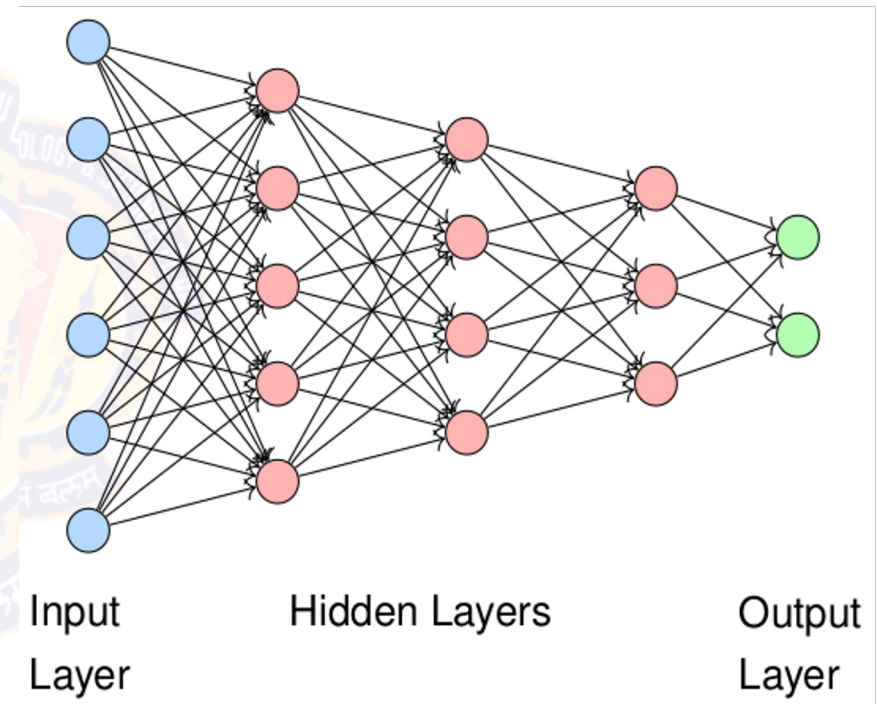
- Neuron is a processing element inspired by how the brain works.
- Similar to biological neuron, each artificial neuron will be do some computation. Each neuron is interconnected to other neurons.
- Similar to brain, the interconnections between neurons store the knowledge it learns. The knowledge is stored as parameters.



What are Artificial Neural Networks?

lead

- Neural Network is a computational model inspired by human brain
- Interconnected neurons process information
- Interconnection can be forward, recursive or self loop
- Learn patterns from data through training
- Capable of approximating complex functions





Properties of Artificial Neural Nets (ANNs)

- Many neuron-like threshold switching units.
- Many weighted interconnections among units.
- Highly parallel, distributed process.
- Emphasis on tuning parameters or weights automatically.



When to consider Neural Networks?

- Input is high-dimensional discrete or real-valued (e.g. raw sensor input).
- Possibly noisy data. Data has lots of errors.
- Output is discrete or real valued or a vector of values.
- Form of target function is unknown.
- Human readability, in other words, explainability, of result is unimportant.
- Examples:
 - Speech phoneme recognition
 - Image classification
 - Financial prediction

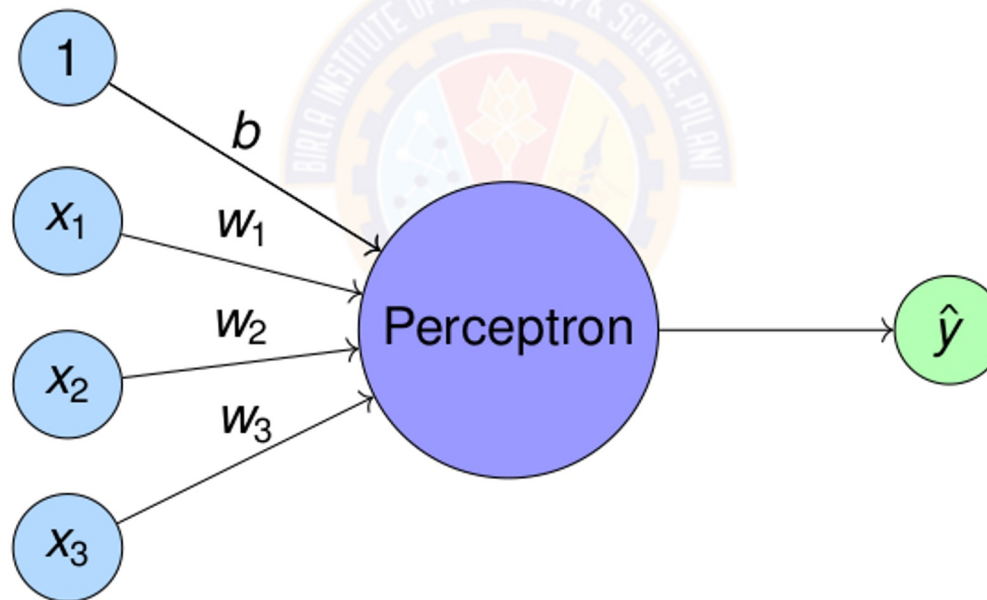


Perceptron

Perceptron

lead

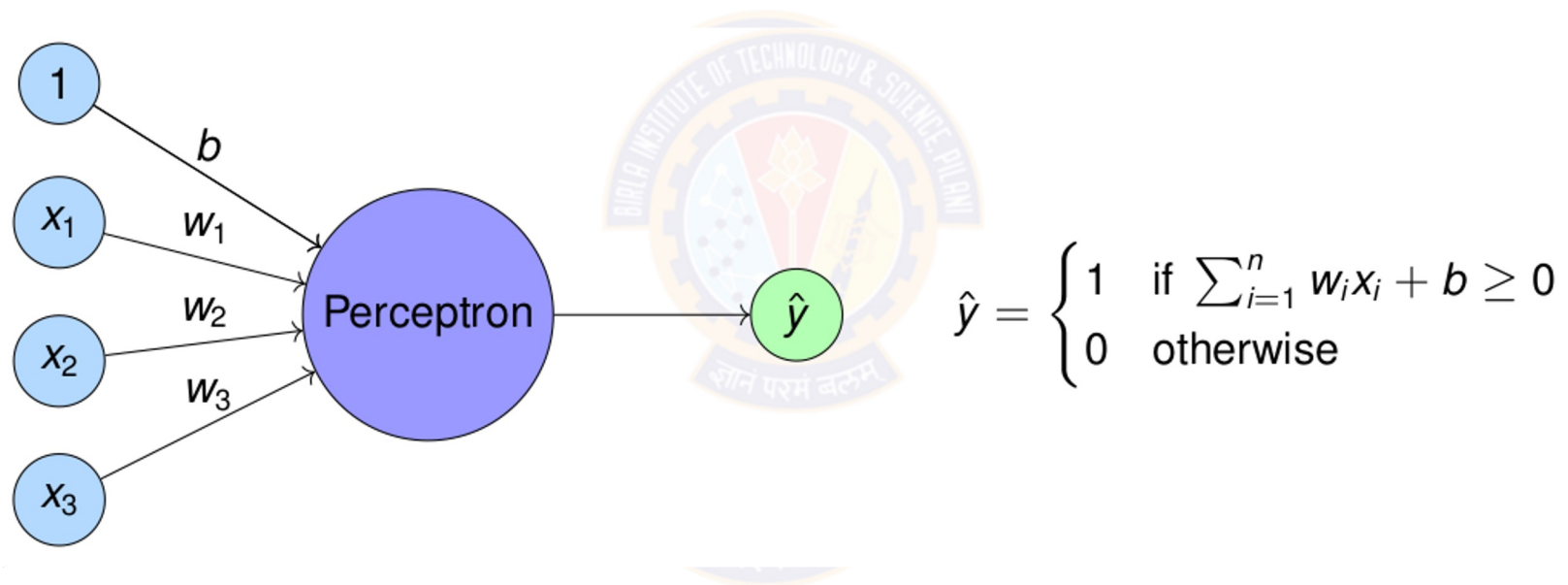
- Perceptron = Simplest form of artificial neuron
- Perceptron was introduced by Rosenblatt in 1958.



Perceptron

lead

- Perceptron = Simplest form of artificial neuron





Representing Logic Gates using Perceptron

- Perceptron can represent all of the primitive Boolean functions AND, OR, NAND and NOR – Linearly separable data
- some Boolean functions cannot be represented by a single perceptron, such as the XOR function – Linearly non separable data
- Perceptron learning rule finds a successful weight vector when the training examples are linearly separable, it can fail to converge if the examples are not linearly separable.



Perceptron & Logic Gates

Perceptron

lead

- Any linear decision can be represented by a Boolean equation.
- The quest is to find out how to represent the above using Perceptron.
- For this we test whether each of the logic gates can be represented by a Perceptron.



AND Logic Gate

lead

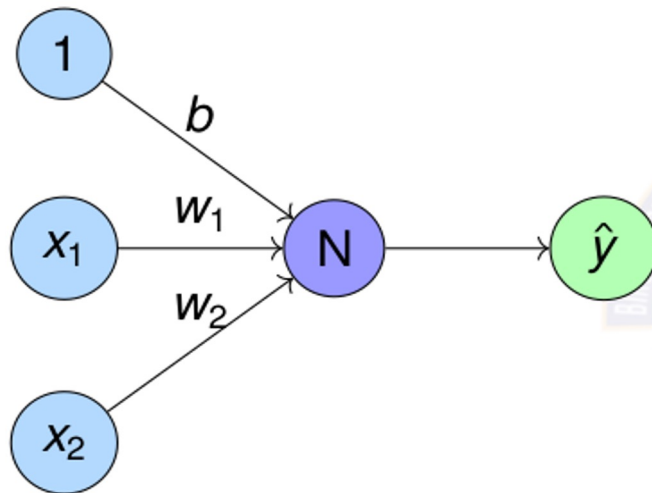
Question:

- How to represent AND gate using a perceptron?
- What are the parameters for the AND perceptron?
- Data is given below.

A	B	A and B
0	0	0
0	1	0
1	0	0
1	1	1

Perceptron for AND gate

lead



Goal: Find parameters w_0, w_1, w_2

x_1	x_2	t	h
0	0	0	$w_0 + 0 + 0 < 0$
0	1	0	$w_0 + 0 + w_2 < 0$
1	0	0	$w_0 + w_1 + 0 < 0$
1	1	1	$w_0 + w_1 + w_2 \geq 0$

One solution is

Perceptron equation is

$$\hat{y} = w_0 x_0 + w_1 x_1 + w_2 x_2$$

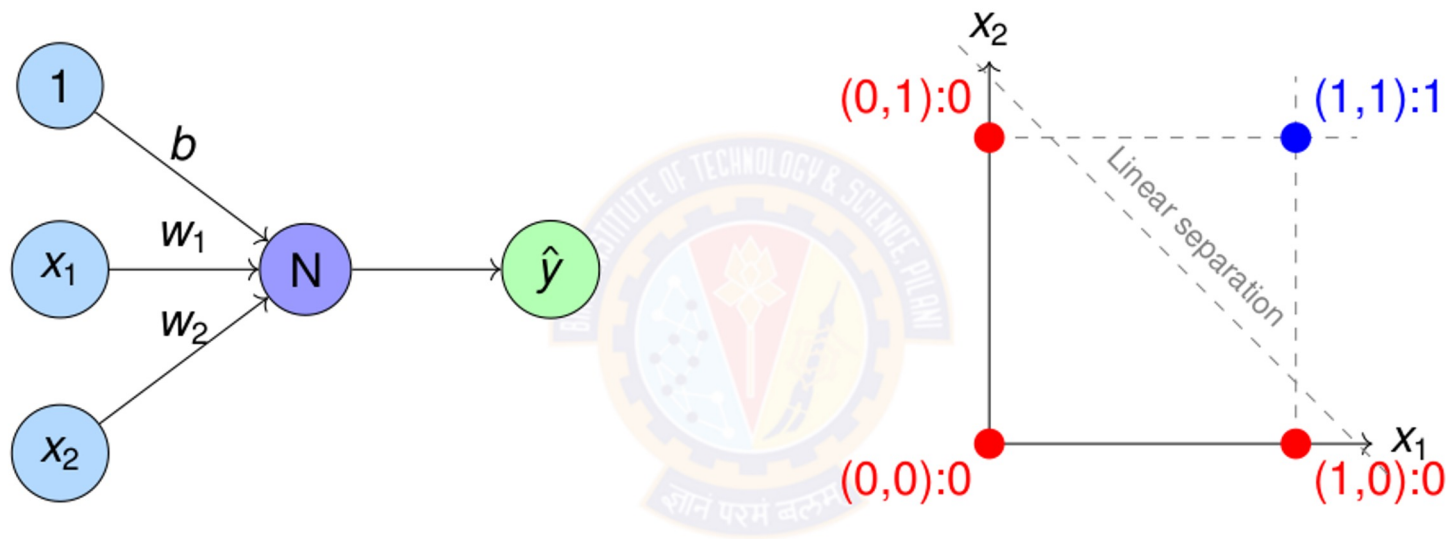
$$w_0 = -1$$

$$w_1 = 0.75$$

$$w_2 = 0.75$$

Perceptron for AND gate

lead



Perceptron equation is

$$\hat{y} = w_0 x_0 + w_1 x_1 + w_2 x_2$$

One solution is

$$w_0 = -1$$

$$w_1 = 0.75$$

$$w_2 = 0.75$$

OR Logic Gate

lead

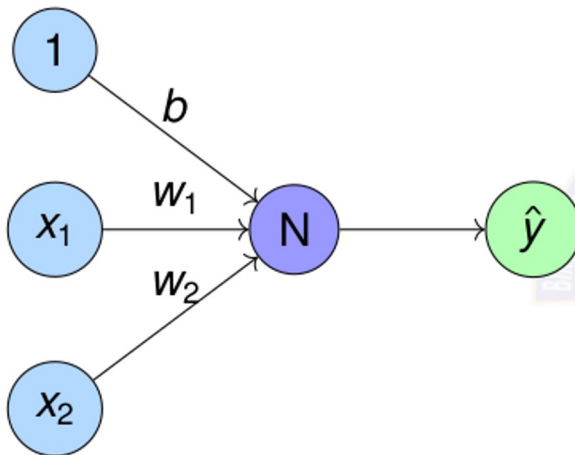
Question:

- How to represent OR gate using a perceptron?
- What are the parameters for the OR perceptron?
- Data is given below.

A	B	A or B
0	0	0
0	1	1
1	0	1
1	1	1

Perceptron for OR gate

lead



Goal: Find parameters w_0, w_1, w_2

x_1	x_2	t	h
0	0	0	$w_0 + 0 + 0 < 0$
0	1	1	$w_0 + 0 + w_2 \geq 0$
1	0	1	$w_0 + w_1 + 0 \geq 0$
1	1	1	$w_0 + w_1 + w_2 \geq 0$

One solution is

Perceptron equation is

$$\hat{y} = w_0 x_0 + w_1 x_1 + w_2 x_2$$

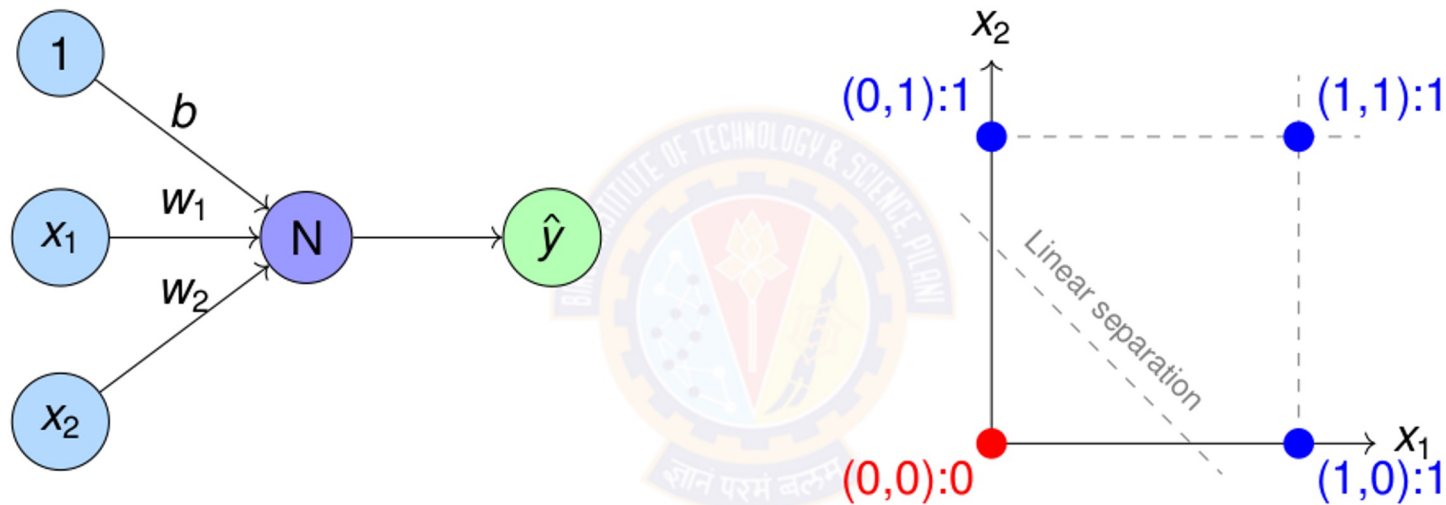
$$w_0 = -1$$

$$w_1 = 2$$

$$w_2 = 2$$

Perceptron for OR gate

lead



Perceptron equation is

$$\hat{y} = w_0 x_0 + w_1 x_1 + w_2 x_2$$

One solution is $w_0 = -1$

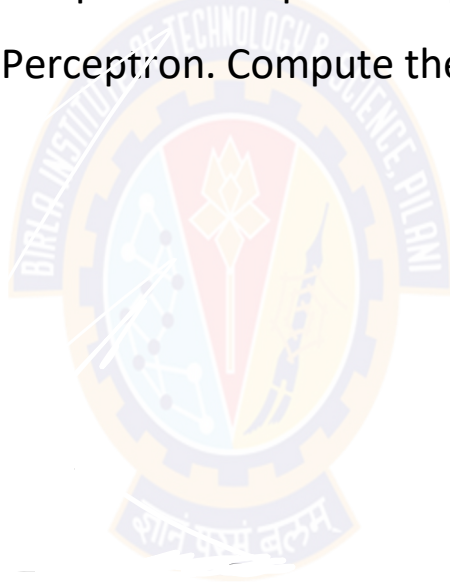
$$w_1 = 2$$

$$w_2 = 2$$

Exercise

lead

1. Represent OR gate using Perceptron. Compute the parameters of the perceptron.
2. Represent NOR gate using Perceptron. Compute the parameters of the perceptron.
3. Represent NAND gate using Perceptron. Compute the parameters of the perceptron.





Perceptron Learning Algorithm

Perceptron Learning Algorithm

lead

Goal: Find weights that correctly classify training data (input, output) pair (x, t) .

Algorithm:

- ① Initialize learning rate $\eta = 0.1$
- ② Initialize weights randomly: $w_i = \text{random}$
- ③ For each training example (x, t) :
 - ▶ Calculate output: $\hat{y} = \text{sign}(\sum_i w_i x_i + b)$
 - ▶ If $\hat{y} \neq t$, update weights:

$$w_i^{\text{new}} = w_i^{\text{old}} + \eta(t - \hat{y})x_i$$

$$b^{\text{new}} = b^{\text{old}} + \eta(t - \hat{y})$$

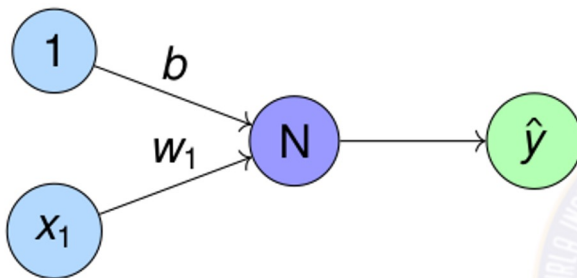
- ④ Repeat until convergence or max iterations

Convergence Theorem:

If linearly separable data, Perceptron learning algorithm will converge in finite steps.

Perceptron Learning for NOT gate

lead

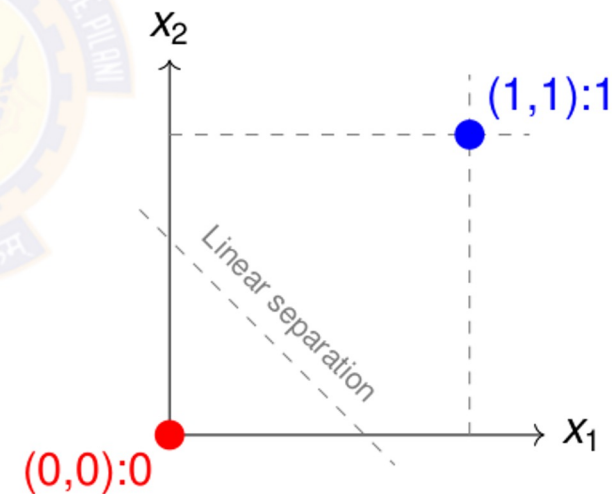


Perceptron equation

$$h = \text{sign}(w_0x_0 + w_1x_1)$$

$$\hat{y} = \begin{cases} 1 & \text{if } h \geq 0 \\ -1 & \text{if } h < 0 \end{cases}$$

Example	x_1	t
Eg:1	-1	1
Eg:2	1	-1



Perceptron Learning Algorithm for NOT gate

- Assume $w_0 = w_1 = 0$. Let the learning rate $= \eta = 1$.
- Equations are

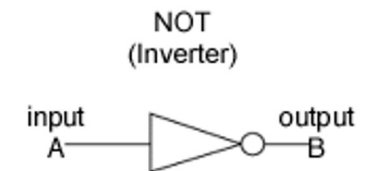
Hypothesis $z = w_0 + w_1 x_1$

Activation $h = \text{sign}(z)$

Compute output $\hat{y} = h = \text{sign}(w_0 + w_1 x_1)$

Compute $\Delta w = \eta(t - \hat{y})x$

Update $w_{\text{new}} \leftarrow w_{\text{old}} + \Delta w$



A	B
0	1
1	0

Perceptron Learning Algorithm for NOT gate

$w_1=w_0=0, \eta =1$

X1	t	W1	W0	z	h	Is equal(t,h)	Δw	New w
0	1	0	0	$0+0=0$	1	YES	No update	$w_1=0$ $w_0=0$
1	0	0	0	$0+0=0$	1	NO	$\Delta w_1=1(0-1)1=-1$ $\Delta w_0=1(0-1)=-1$	$W_1=0-1=-1$ $w_0=0-1=-1$

Epoch-2 for NOT gate



$w1=-1, w0=-1, \eta =1$

X1	t	W1	W0	z	h	lsequal(t,h)	Δw	New w
0	1	-1	-1	$0-1=-1$	0	NO	$\Delta w1=1(1-0)0=0$ $\Delta w0=1(1-0) =1$	$W1= -1+0=-1$ $w0=-1+1=0$
1	0	-1	0	$-1+0=-1$	0	yes	No update	$W1= -1$ $w0=0$

Epoch-3 for NOT gate

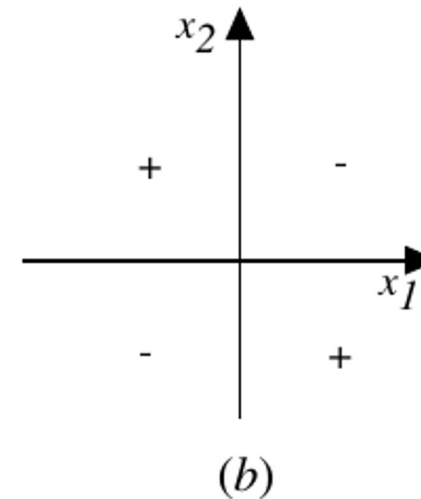
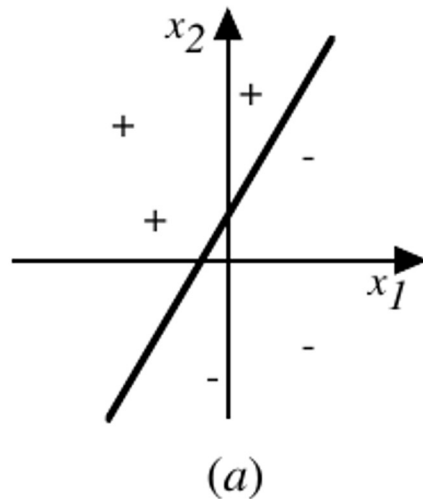
$w1=-1, w0=0, \eta =1$

X1	t	W1	W0	z	h	lsequal(t,h)	Δw	New w
0	1	-1	0	$0+0=0$	1	yes	No update	W1= -1 w0=0
1	0	-1	0	$-1+0=-1$	0	yes	No update	W1= -1 w0=0

Representational Power of Perceptrons



- A perceptron represents a hyperplane decision surface in the n -dimensional space of examples.
- The perceptron outputs a 1 for examples lying on one side of the hyperplane and outputs a -1 for examples lying on the other side.



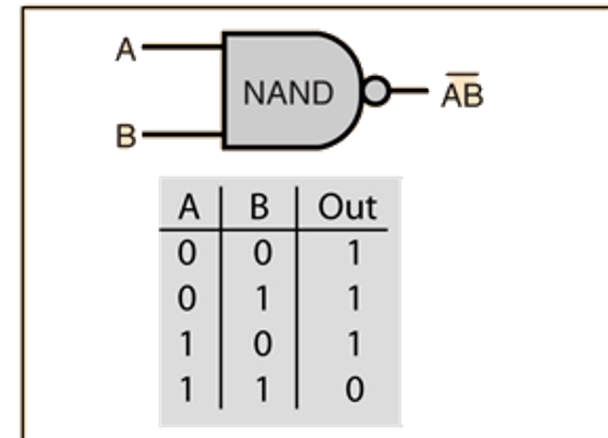
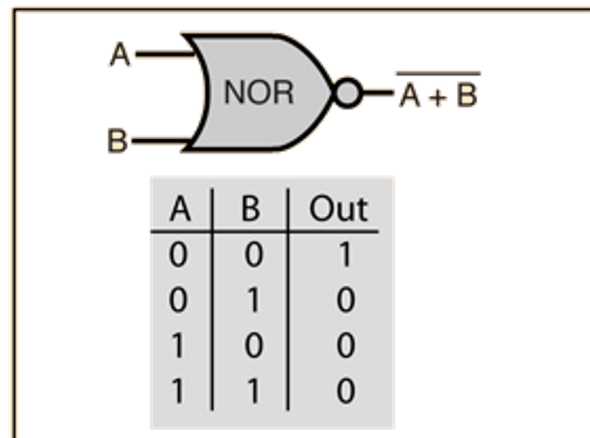
Exercise



Represent following gates using Perceptron. Compute the parameters of the perceptron.



A	B	AB
0	0	0
0	1	1
1	0	1
1	1	1





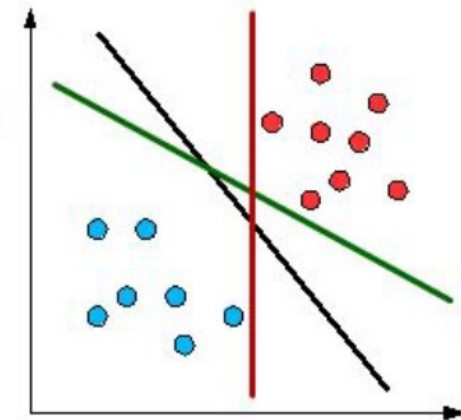
Single Layer Perceptron

Single Layer Perceptron (SLP)



Linearly Separable Data

- Two sets of data points in a two dimensional space are said to be linearly separable when they can be completely separable by a single straight line.
- In general, **two groups of data points are separable in a n -dimensional space if they can be separated by an $(n-1)$ dimensional hyperplane.**
- A straight line can be drawn to separate all the data examples belonging to class +1 from all the examples belonging to the class -1. Then the two-dimensional data are clearly linearly separable.
- An infinite number of straight lines can be drawn to separate the class +1 from the class -1.

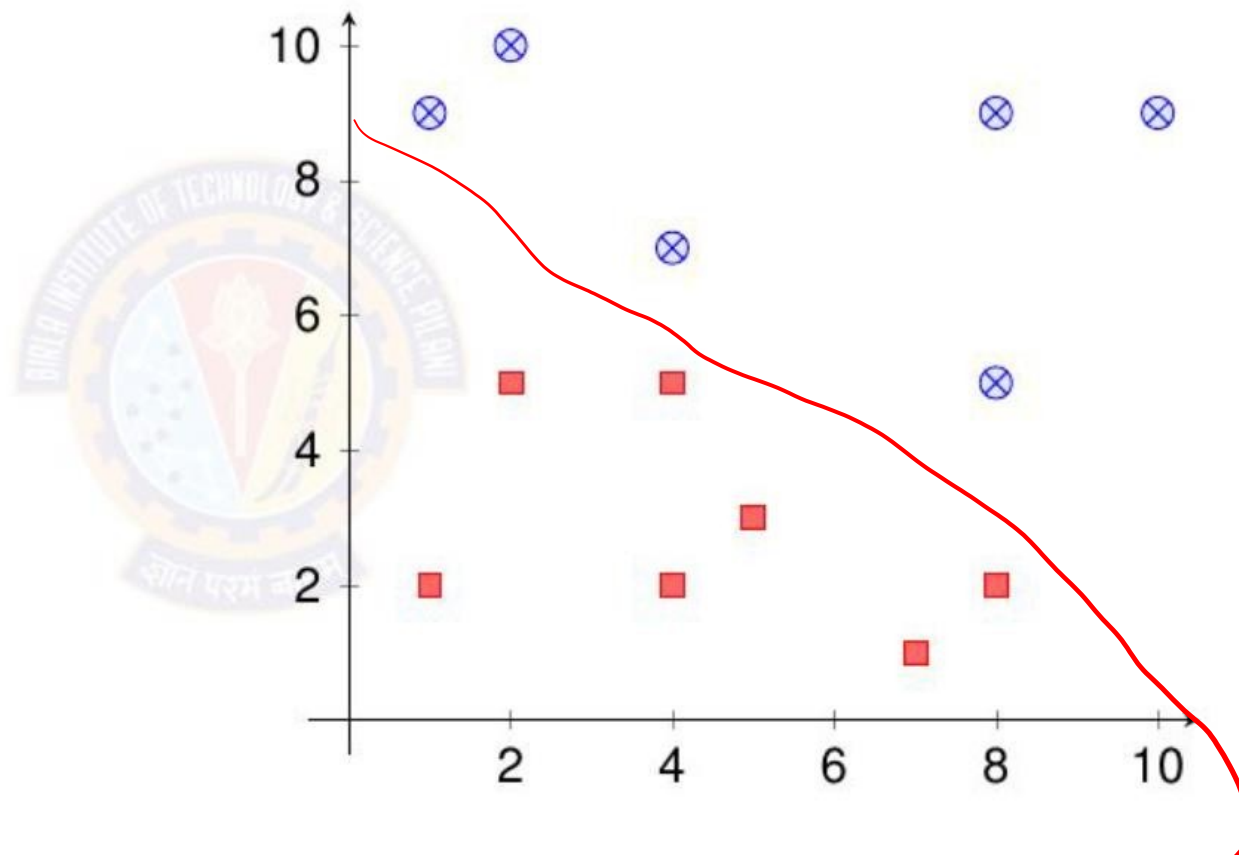


Single Layer Perceptron (SLP)



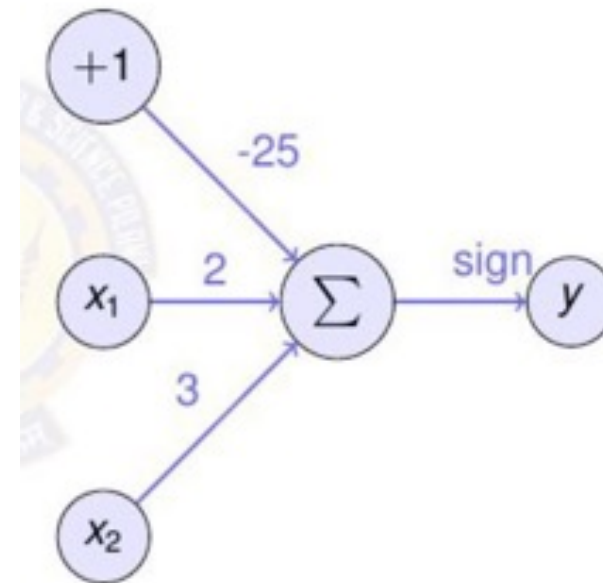
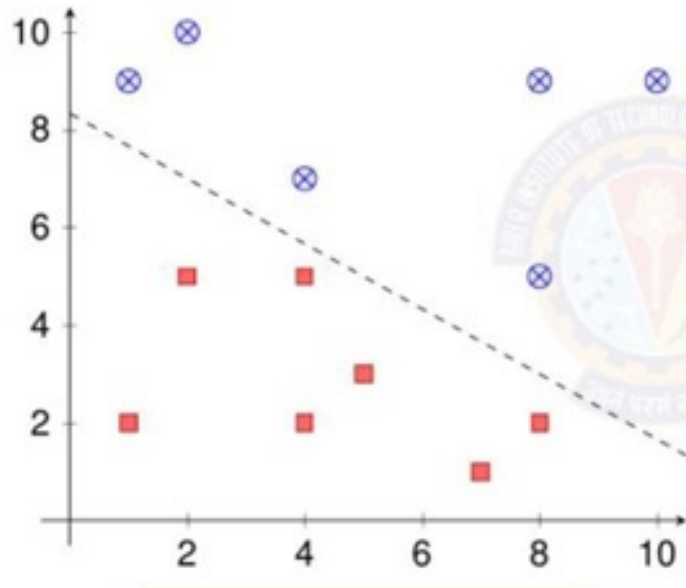
Example of Linearly Separable data

x_1	x_2	y
1	9	white
10	9	white
4	7	white
4	5	pink
5	3	pink
8	9	white
4	2	pink
2	5	pink
7	1	pink
2	10	white
8	5	white
1	2	pink
8	2	pink



Single Layer Perceptron (SLP)

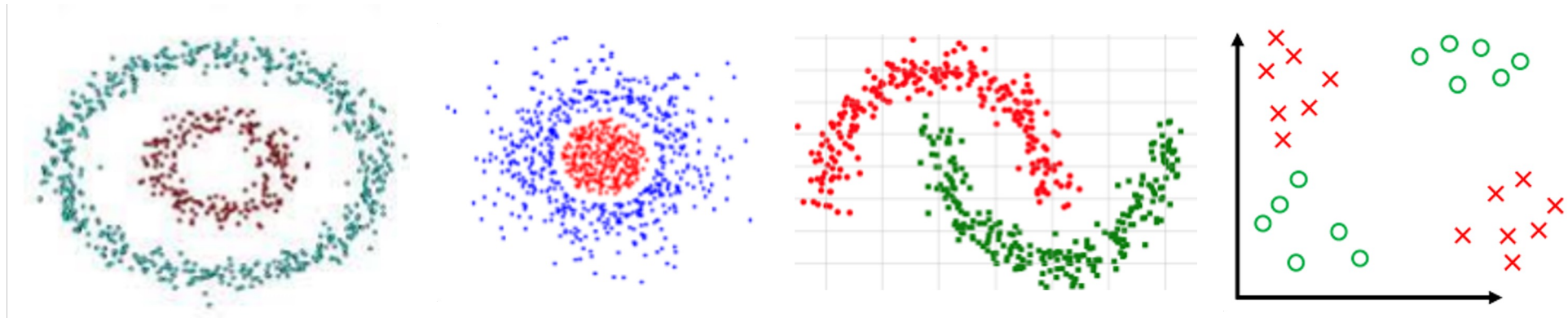
- Equation of a decision surface $x_2 = mx_1 + c$
- Or more generally, it can also be written as: $\mathbf{ax}_1 + \mathbf{bx}_2 + \mathbf{d} = 0$,
- where slope (m) = $-a/b$ and x_2 -axis intercept (c) = $-d/b$
- E.g $2x_1 + 3x_2 - 25 = 0$



Non-linearly Separable Data



- Two groups of data points are non-linearly separable in a 2-dimensional space if they cannot be easily separated with a linear line.

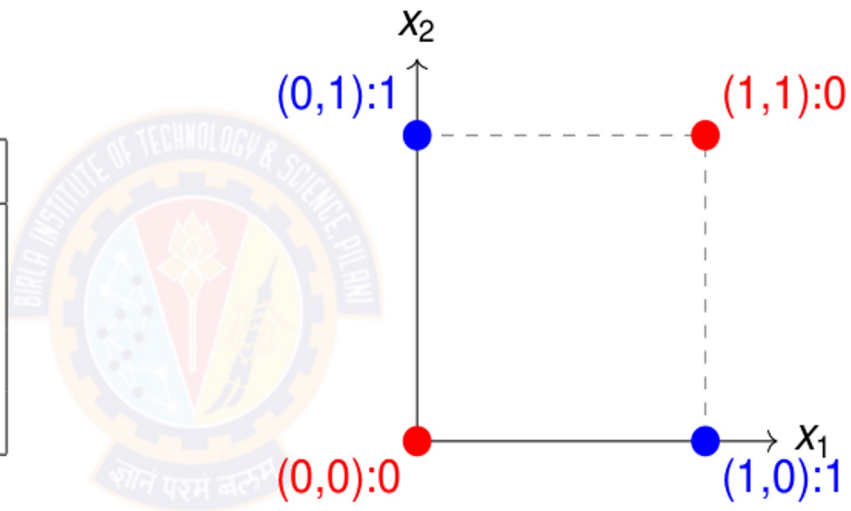


The XOR Problem

lead

Truth Table for XOR:

x_1	x_2	XOR Output
0	0	0
0	1	1
1	0	1
1	1	0



PROBLEM

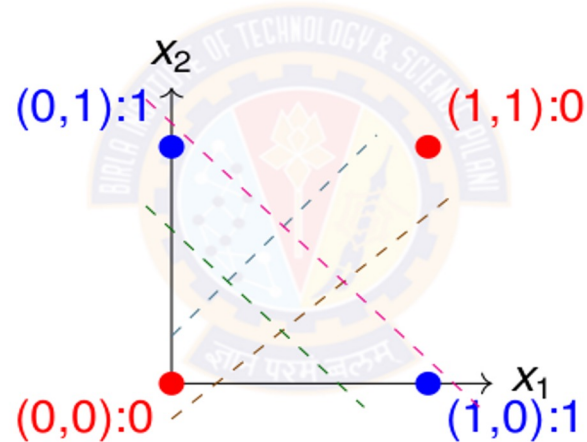
XOR is not linearly separable! Single perceptron cannot solve XOR.

Why Single Perceptron Fails on XOR

lead

Linear Decision Boundary Limitation:

Any line $w_1x_1 + w_2x_2 + b = 0$ cannot separate XOR classes.



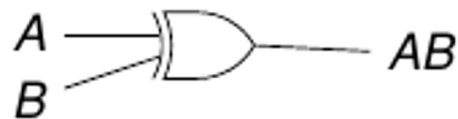
No single line works!

Solution: Need multiple layers (Multilayer Perceptron)

Perceptron for XOR Gate

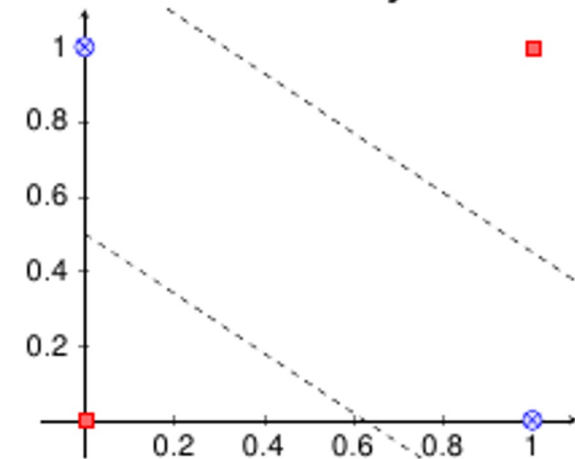
Question:

- How to represent XOR gate using a Perceptron?
- What are the parameters for the XOR Perceptron?
- Data is given below.



A	B	AB
0	0	0
0	1	1
1	0	1
1	1	0

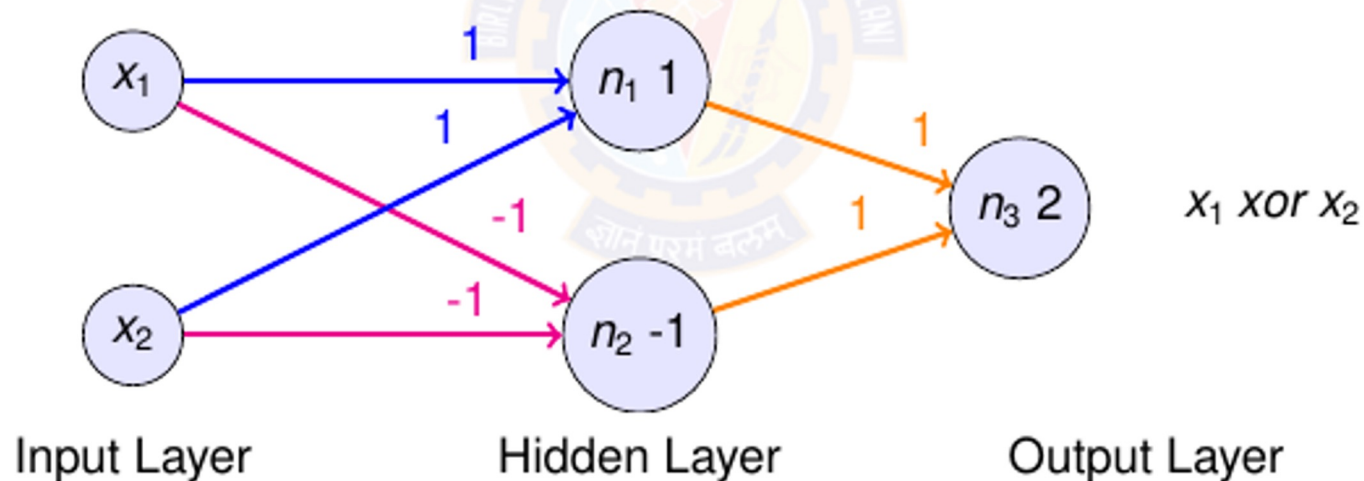
Challenge: Data is non-linearly separable.
Decision boundary for XOR



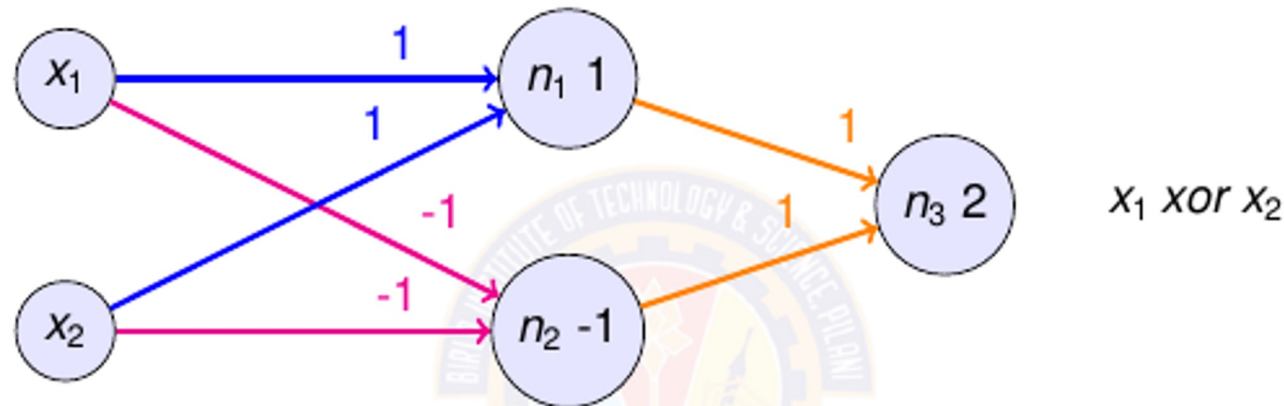
Solution for XOR

Qn: How to represent XOR gate using a Perceptron?

- Use Multilayer Perceptron (MLP)
- Introduce another layer in between the input and output.
- This in-between layer is called hidden layer.



MLP for XOR



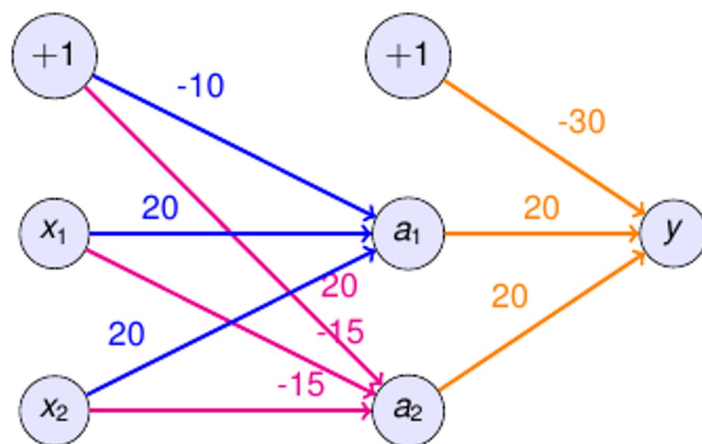
x_1	x_2	n_1	n_2	n_3
0	0	$0 \cdot 1 + 0 \cdot 1 = 0 \not\geq th$ $n_1 = 0$	$0 \cdot (-1) + 0 \cdot (-1) = 0 > th$ $n_2 = 1$	$0 \cdot 1 + 1 \cdot 1 = 1 \not\geq th$ $n_3 = 0$
0	1	$0 \cdot 1 + 1 \cdot 1 = 1 \geq th$ $n_1 = 1$	$0 \cdot (-1) + 1 \cdot (-1) = -1 \geq th$ $n_2 = 1$	$1 \cdot 1 + 1 \cdot 1 = 2 \geq th$ $n_3 = 1$
1	0	$1 \cdot 1 + 0 \cdot 1 = 1 \geq th$ $n_1 = 1$	$1 \cdot (-1) + 0 \cdot (-1) = -1 \geq th$ $n_2 = 1$	$1 \cdot 1 + 1 \cdot 1 = 2 \geq th$ $n_3 = 1$
1	1	$1 \cdot 1 + 1 \cdot 1 = 2 \geq th$ $n_1 = 1$	$1 \cdot (-1) + 1 \cdot (-1) = -2 \not\geq th$ $n_2 = 0$	$1 \cdot 1 + 0 \cdot 1 = 1 \not\geq th$ $n_3 = 0$

- Multi-layer perceptron is a first generic model that refers to any layered network of neurons
- Why called hidden layer, since their output is not directly observed, but is fed into the third neuron

MLP for XOR



An XOR cannot be represented using a single perceptron as the XOR has non-linear decision boundary. To represent non-linear decision boundary we need a multi-layer perceptron with one hidden layers.



$$a_k = \text{sign}(w_{k1}x_1 + w_{k2}x_2 + b)$$

x_1	x_2	a_1	a_2	y
0	0	$-10 \sim 0$	$20 \sim 1$	$-10 \sim 0$
0	1	$10 \sim 1$	$5 \sim 1$	$10 \sim 1$
1	0	$10 \sim 1$	$5 \sim 1$	$10 \sim 1$
1	1	$30 \sim 1$	$-30 \sim 0$	$-10 \sim 0$

Issues of Depth and Width

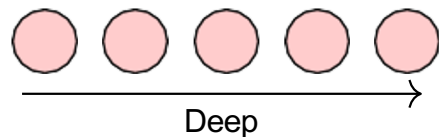


- The architecture of neural networks often specified by the width and the depth of the networks.
- The depth h of a network is defined as its number of layers (including output layer but excluding input layer)
- The width d_m of a network is defined to be the maximal number of nodes in a layer.

NETWORK DEPTH VS WIDTH

Two ways to increase model capacity:

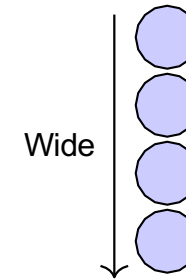
Depth (more layers):



Properties:

- Hierarchical features
- More expressive
- Better generalization

Width (more units per layer):



Properties:

- More parallel features
- Higher capacity
- Risk of overfitting

Issues of Depth and Width



- While depth emphasizes the sequential processing of information, width focuses on parallelization and neuron abundance.
- A deep network can represent complex functions using far fewer neurons than a shallow, wide network.
- **Advantage of DL:** Deep learning excels at analysing and learning from unstructured or complex data, such as images, videos, natural language and audio sequences.
- **Disadvantage of DL:** The black box problem, overfitting, lack of contextual understanding, data requirements, and computational intensity

References



Chapter – 1

<http://mlsp.cs.cmu.edu/people/rsingh/IDLbook.html>



Chapter 4 of Book: Machine Learning by Tom M. Mitchell



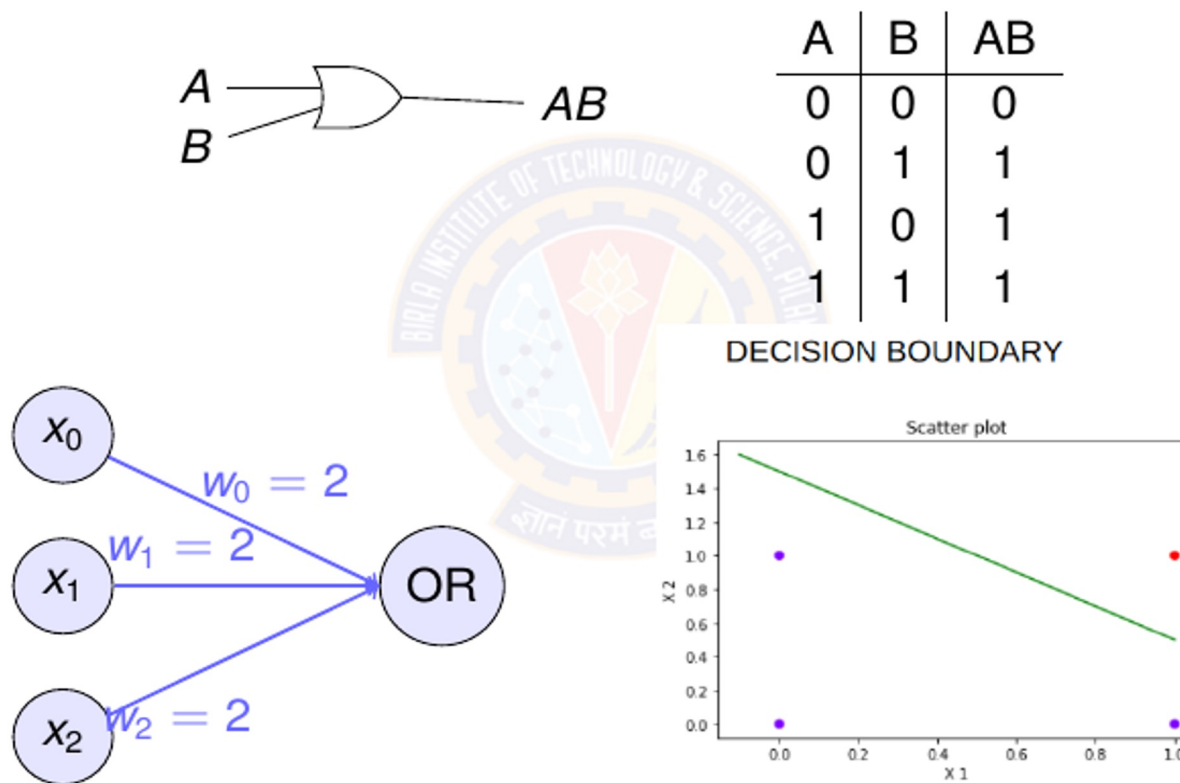
Thank You All !



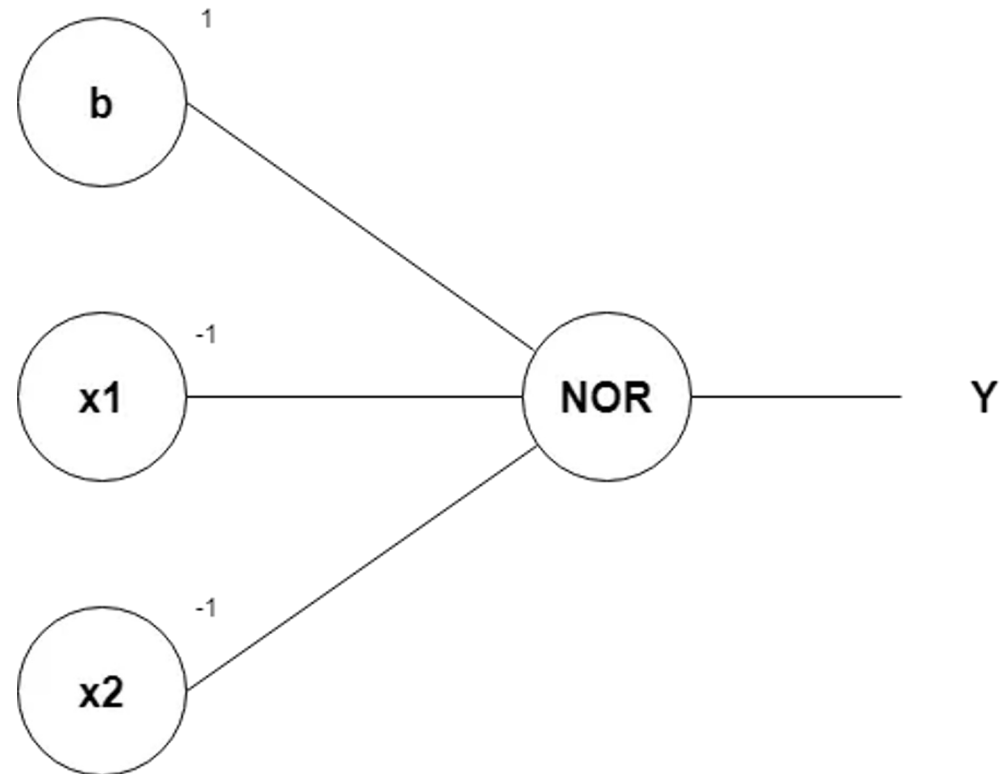
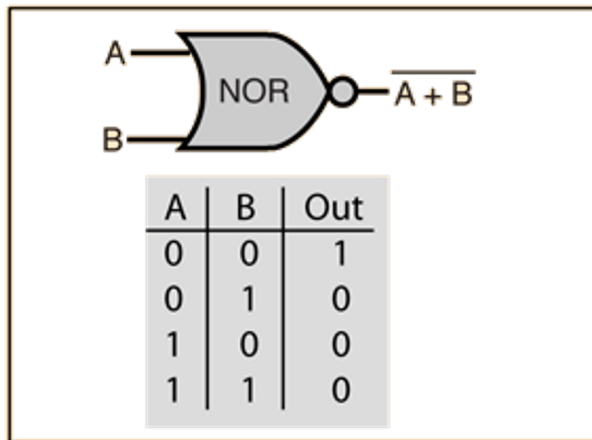
Exercise for practice



Perceptron for OR gate



Perceptron for NOR Gate



Perceptron for NAND Gate

